
ARITHMETIC EXPRESSION GEOMETRY I: FOUNDATIONS

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ABSTRACT

We introduce Arithmetic Expression Geometry (AEG), a framework in which arithmetic expressions are studied through the geometry of their evaluation histories. For a distinguished class of threadlike expressions, additive and multiplicative updates determine two generating directions of propagation, and their continuous limit yields an arithmetic flow equation with an $\text{Aff}(1)$ Lie-theoretic formulation. We define an *Arithmetic Expression Space* (AES) as a Riemannian surface equipped with an assignment field satisfying the corresponding first-order law, and we construct basic model spaces $\mathfrak{E}_0$ and $\mathfrak{E}_1$. The first intrinsic invariant of the theory is *arithmetic torsion*, arising from the non-commutativity of additive and multiplicative propagation. To globalize this invariant, we introduce the *Accumulative Commutative Space* (ACS) and prove a Stokes-type triple identity expressing torsion equivalently as an evaluation difference, a weighted area integral, and a boundary integral.

To localize the same phenomenon, we construct a three-dimensional contact manifold with contact form $\alpha = da - (\mu du + \lambda a dv)$. Its horizontal distribution recovers the propagation directions, yields a natural horizontal differential calculus δ , and produces a commutator curvature package proportional to $\mu\lambda du \wedge dv$. On this horizontal layer we define *horizontal holomorphic fields* via an arithmetic Cauchy–Riemann system and show that it forces a nontrivial twisted second-order equation. Taken together, these results provide a theorem-driven foundational layer of AEG and clarify the normalization and invariance properties of its core structures.

Keywords arithmetic expressions, hyperbolic geometry

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1 Introduction

1.1 Motivation and guiding examples

Arithmetic expressions come with a hidden combinatorial choice: an *evaluation history*. Even when two evaluation histories give the same final value, intermediate values can differ, and commutators of elementary steps can carry systematic defects. Arithmetic Expression Geometry (AEG) is a proposal to treat such defects as geometric data.

The present paper focuses on a distinguished class of *threadlike* expressions, where evaluation histories naturally decompose into two elementary propagations: additive updates and multiplicative updates. Passing from discrete propagation to a continuous limit yields an arithmetic flow equation. The central theme is that the non-commutativity between these two propagations admits both global invariants and local differential-geometric incarnations.

1.2 Main results and structure of the paper

The paper establishes a first foundational layer of AEG. Its contributions are organized into the following theorem-level packages.

- *Arithmetic flow equation.* From infinitesimal additive and multiplicative propagations we derive a first-order propagation law and reformulate it in a Lie-theoretic $\text{Aff}(1)$ framework.
- *Arithmetic Expression Spaces (AES).* We define an AES as geometric data supporting an assignment field governed by the flow/eikonal law, and we construct the basic model spaces $\mathfrak{E}_0$ and $\mathfrak{E}_1$, including verifiable computations in the upper half-plane.
- *Arithmetic torsion and the ACS triple identity.* We define arithmetic torsion as a commutator defect of additive and multiplicative propagation. We introduce the Accumulative Commutative Space (ACS) and prove a Stokes-type triple identity expressing torsion equivalently as an evaluation difference, a weighted area integral, and a boundary integral under explicit hypotheses.
- *Contact structure and horizontal calculus.* We lift the propagation data to a contact manifold whose horizontal distribution recovers the flow directions. This yields a horizontal differential calculus δ and a local commutator curvature package reflecting the same arithmetic non-commutativity.
- *Horizontal holomorphic fields and twisted harmonicity.* On the horizontal distribution we define an arithmetic Cauchy–Riemann system and package it into first-order operators. We show that this first-order system forces a nontrivial twisted second-order equation, providing an intrinsic analytic operator layer associated to arithmetic propagation.
- *Normalization and invariance.* We systematize the effects of rescaling, change of logarithmic base, and coordinate changes, and we record the invariance properties of the core constructions and identities.

The remainder of the paper follows this structure. Sections 2–3 develop the propagation law and the definition and construction of AES. Section 4 develops arithmetic torsion and the ACS triple identity. Section 5 develops the contact-geometric lift and the horizontal calculus. Section 6 introduces horizontal holomorphic fields and their induced twisted harmonicity. Section 7 collects normalization, invariance, and mathematical positioning. Technical computations and examples are placed in the appendices.

1.3 What is established here and what is deferred

This paper is foundational and intentionally delimits its scope.

- We give explicit definitions, model constructions, and complete proofs for the central identities claimed in the main text.
- We do not attempt a global classification of AES beyond the model settings treated here. Classification and uniqueness problems are stated explicitly as open directions.
- The horizontal holomorphic and twisted harmonic operators introduced here are not presented as a full replacement of classical complex analysis. In particular, boundary integral kernels, analytic continuation, and global potential theory are deferred to later work.
- The multi-layered equality program and the associated loop/relations theory are only motivated by examples in this paper; a systematic global theory is deferred.

1.4 Relation to existing frameworks

Several parts of the construction naturally interface with established geometric structures. The contact form and horizontal calculus invite comparison with jet-space contact geometry and with CR/sub-Riemannian analysis on non-integrable distributions. From a combinatorial viewpoint, evaluation histories and their commutators suggest a groupoid language for rewriting systems and a holonomy-type organization of equality levels. Section 7 records these comparisons at the level needed for the present foundational development.

2 Arithmetic paths and the flow equation

2.1 Threadlike expressions and propagation order

We begin with arithmetic expressions over the rational numbers $\mathbb{Q}$, in order to keep the first layer of the theory combinatorial and to avoid analytic subtleties unrelated to evaluation histories.

Definition 2.1. An arithmetic expression over $\mathbb{Q}$ is generated by the production rules

$$\begin{aligned} a &\longleftarrow x, \\ a &\longleftarrow (a + a), \\ a &\longleftarrow (a - a), \\ a &\longleftarrow (a \times a), \\ a &\longleftarrow (a \div a), \end{aligned} \tag{1}$$

where $x \in \mathbb{Q}$. We write $a \in \mathbb{E}[\mathbb{Q}]$.

An expression may be viewed either as a formal string or as a syntax tree whose leaves are numbers and whose internal nodes are the operators $+$, $-$, $\times$, $\div$. Evaluation is the usual recursive process, and it is a partial function

$$\nu : \mathbb{E}[\mathbb{Q}] \rightarrow \mathbb{Q},$$

undefined exactly when a division by zero occurs during evaluation. Unless explicitly stated otherwise, expressions in this paper are assumed to be *evaluable*, i.e. $\nu(a)$ is defined.

Definition 2.2. The evaluation order of an arithmetic expression is an ordering of the internal nodes in its syntax tree such that each node is evaluated only after all of its children have been evaluated.

Even for elementary expressions, evaluation order need not be unique. AEG is driven by the observation that a fixed endpoint value may conceal several distinct evaluation histories.

We isolate a particularly rigid class for which evaluation histories behave like directed paths.

Definition 2.3. A threadlike expression is an arithmetic expression whose tree representation has the property that every left child is a leaf.

Intuitively, a threadlike expression is built by starting from one initial number and repeatedly attaching a new operation and a new number on the right. Such expressions admit a canonical left-to-right reading which is convenient for geometry.

Currying and path notation

We rewrite threadlike expressions as compositions of unary operators acting on the current value. We use the basic operators

- $\oplus_y : x \mapsto x + y$,
- $\ominus_y : x \mapsto x - y$,
- $\otimes_y : x \mapsto x \cdot e^y$,
- $\oslash_y : x \mapsto x \cdot e^{-y}$,

where $y \in \mathbb{R}$ records the additive increment or logarithmic multiplicative increment.

For operators $a_1, a_2, \dots, a_n$, we write

$$xa_1a_2 \cdots a_n := a_n(a_{n-1}(\cdots a_2(a_1(x)) \cdots)).$$

If a path begins with a number, we call it a *bounded path*; if it consists only of operators, we call it a *free path*. A bounded path evaluates to a number, while a free path evaluates to a function.

Lemma 2.1. The path notation is associative: for paths a, b, c we have $a[bc] = [ab]c$.

Proof. The statement is a direct unraveling of function composition for bounded and free paths. □

Definition 2.4. The concatenation of two paths p_1 and p_2 is their composite as functions:

$$p_1 \cdot p_2 := p_2 \circ p_1.$$

A category and a groupoid of paths

The preceding language admits a convenient categorical packaging which will later organize equality levels and loop phenomena.

Let X be a set of admissible values (for example $X = \mathbb{R}$, or a subset on which all operators under consideration are defined). Let $\mathcal{P}(X)$ be the category whose objects are elements of X and whose morphisms are bounded paths: a morphism $x \rightarrow y$ is a finite sequence of operators p such that $y = p(x)$. Composition is concatenation of paths, and the identity morphism at x is the empty path.

When the operators allowed in paths are invertible, $\mathcal{P}(X)$ is a groupoid. In particular, the operators $\oplus_y, \ominus_y, \otimes_y, \oslash_y$ are invertible, and the inverse of a bounded path is obtained by reversing the sequence and replacing each operator by its inverse.

Free paths can be viewed as the arrow labels of an *action groupoid*. Let $\mathcal{W}$ denote the set of all free paths built from the chosen operators, with multiplication given by concatenation. Each $p \in \mathcal{W}$ acts (partially, if desired) on X by evaluation. The corresponding action groupoid has objects X and morphisms represented by pairs (p, x) with source x and target $p(x)$, modulo the obvious identification of different words inducing the same function on their common domain.

This viewpoint separates two kinds of information:

- a *morphism* $x \rightarrow y$, which records the full evaluation history as an arrow in a path groupoid;
- an *endpoint relation* between x and y , obtained by forgetting which arrow was used to connect them.

Loops are endomorphisms $x \rightarrow x$ in this groupoid. They encode relations among histories and will later support the multi-layered equality program: two histories may agree as arrows, may become equal after imposing relations among loops, or may merely share the same endpoints while remaining distinct arrows.

2.2 Additive and multiplicative infinitesimal propagations

Among threadlike expressions, the most useful bookkeeping class is the alternating one: a zigzag of multiplication and addition steps. Concretely, an alternating threadlike expression is a word of the form

$$\alpha = a_1 b_1 a_2 b_2 \cdots a_l b_l, \quad a_i = \otimes_{\lambda_i}, \quad b_i = \oplus_{\mu_i}, \quad \lambda_i, \mu_i \in \mathbb{R}. \quad (2)$$

It is naturally a free path, and it becomes a bounded path once an initial value is attached.

To pass from discrete histories to a differential law, we now freeze two real parameters $\mu, \lambda \in \mathbb{R}$ and isolate the two elementary generators

- $\oplus_\mu : x \mapsto x + \mu$,
- $\ominus_\mu : x \mapsto x - \mu$,
- $\otimes_\lambda : x \mapsto x \cdot e^\lambda$,
- $\oslash_\lambda : x \mapsto x \cdot e^{-\lambda}$.

These are the infinitesimal prototypes of “add/subtract” and “multiply/divide”. Their defining feature is that the additive action is independent of the current value while the multiplicative action scales the current value.

The first structural invariant appears already at this discrete level: additive and multiplicative propagation do not commute. Acting on a value x , we have the commutator defect

$$\tau(x) := x \oplus_\mu \otimes_\lambda - x \otimes_\lambda \oplus_\mu = \mu(e^\lambda - 1). \quad (3)$$

We call τ the *arithmetic torsion*. Later sections reinterpret this commutator defect globally in the ACS and locally through contact geometry; here we record it only as a guide for what the differential limit must remember.

2.3 Derivation of the arithmetic flow equation

We now idealize the two generators as infinitesimal motions in two orthogonal directions. Let (u, v) be a local orthogonal coordinate system in which the u -direction represents the additive generator and the v -direction represents the multiplicative generator. We interpret μ and λ as the infinitesimal intensities of these two actions.

Let s denote arc length along a unit-speed path and let θ be the angle that the tangent vector makes with the positive u -direction, so that

$$du = \cos \theta ds, \quad dv = \sin \theta ds.$$

Let $a = a(s)$ be the current assignment value. Over an infinitesimal step ds , the additive generator contributes μdu while the multiplicative generator contributes the factor $e^{\lambda dv}$. There are two natural orderings of these two actions:

$$a(s + ds) = (a + \mu du)e^{\lambda dv} + O(ds^2) \quad \text{or} \quad a(s + ds) = ae^{\lambda dv} + \mu du + O(ds^2).$$

Their difference is second order, in accordance with the commutator defect (3). Expanding $e^{\lambda dv} = 1 + \lambda dv + O(ds^2)$ gives

$$a(s + ds) = a + \mu du + a\lambda dv + O(ds^2) = a + (\mu \cos \theta + a\lambda \sin \theta)ds + O(ds^2).$$

Dividing by ds and passing to the limit yields the fundamental differential relation.

Proposition 2.1 (Arithmetic flow equation). *Along a unit-speed path whose tangent makes angle θ with the additive direction, the assignment value a satisfies*

$$\frac{da}{ds} = \mu \cos \theta + a\lambda \sin \theta. \quad (4)$$

For a fixed direction θ , equation (4) is a linear first-order ODE. In the axis directions $\theta \in \frac{\pi}{2}\mathbb{Z}$ it reproduces the four elementary motions:

- $\theta = 0$: $a(s) = a_0 + \mu s$,
- $\theta = \frac{\pi}{2}$: $a(s) = a_0 e^{\lambda s}$,
- $\theta = \pi$: $a(s) = a_0 - \mu s$,
- $\theta = \frac{3\pi}{2}$: $a(s) = a_0 e^{-\lambda s}$.

In this sense, the flow equation is the infinitesimal envelope of the discrete generator calculus encoded by $\oplus_\mu$ and $\otimes_\lambda$.

2.4 Lie-theoretic formulation via $\text{Aff}(1)$

The flow equation admits a convenient Lie-theoretic encoding which separates the group-level propagation from its infinitesimal generator. We use the following notational convention throughout: a Roman letter such as G denotes a Lie group, while the corresponding fraktur letter such as $\mathfrak{g}$ denotes its Lie algebra.

The affine group and its Lie algebra

Let

$$G := \text{Aff}(1) = \left\{ \begin{pmatrix} \Phi & a \\ 0 & 1 \end{pmatrix} \mid \Phi \in \mathbb{R}_{>0}, a \in \mathbb{R} \right\},$$

acting on $x \in \mathbb{R}$ by $x \mapsto \Phi x + a$. The group law is matrix multiplication, equivalently

$$(\Phi_1, a_1)(\Phi_2, a_2) = (\Phi_1 \Phi_2, a_1 + \Phi_1 a_2).$$

This semidirect-product rule already mirrors propagation: a multiplicative scaling Φ_1 amplifies a subsequent additive translation a_2 .

Its Lie algebra is

$$\mathfrak{g} := \mathfrak{aff}(1) = \left\{ \begin{pmatrix} \alpha & \beta \\ 0 & 0 \end{pmatrix} \mid \alpha, \beta \in \mathbb{R} \right\},$$

with bracket given by the matrix commutator. A convenient basis is

$$H := \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad N := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad [H, N] = N.$$

The one-parameter subgroups $\exp(tH) = \begin{pmatrix} e^t & 0 \\ 0 & 1 \end{pmatrix}$ and $\exp(tN) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ encode, respectively, pure multiplication and pure addition. In particular,

$$\rho(\otimes_\lambda) = \exp(\lambda H), \quad \rho(\oplus_\mu) = \exp(\mu N),$$

matching the discrete generators used in §2.2.

More generally, any free path p built from these generators determines (when it is defined) an affine transformation of the line, hence an element $\rho(p) \in G$. The resulting map ρ is a concrete realization of the “word $\rightarrow$ operator” passage implicit in the path notation. The differential viewpoint corresponds to passing from short words to their infinitesimal generators: for example, $\exp(\varepsilon H)$ and $\exp(\varepsilon N)$ encode the elementary multiplicative and additive updates to first order in ε , and the commutator defect recorded in (3) appears as a second-order obstruction to commutativity.

A left-invariant evolution and the flow equation

Fix a direction θ . We prescribe the Lie-algebra element

$$\Omega(\theta) := \lambda \sin \theta H + \mu \cos \theta N = \begin{pmatrix} \lambda \sin \theta & \mu \cos \theta \\ 0 & 0 \end{pmatrix}. \quad (5)$$

Consider the left-invariant linear system on G ,

$$\frac{d}{ds}g(s) = \Omega(\theta) g(s), \quad g(s) \in G. \quad (6)$$

Writing

$$g(s) = \begin{pmatrix} \Phi(s) & a(s) \\ 0 & 1 \end{pmatrix},$$

equation (6) becomes

$$\begin{pmatrix} \Phi'(s) & a'(s) \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} \lambda \sin \theta & \mu \cos \theta \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \Phi(s) & a(s) \\ 0 & 1 \end{pmatrix}.$$

Comparing entries yields

$$\Phi'(s) = \lambda \sin \theta \Phi(s), \quad a'(s) = \mu \cos \theta + a(s) \lambda \sin \theta.$$

The second equation is exactly Proposition 2.1. Thus the assignment component of the arithmetic flow is the translation component of a left-invariant evolution on the affine group.

3 Arithmetic Expression Spaces

3.1 Definition of an Arithmetic Expression Space

We now package the preceding discussion into the basic object of the paper.

Definition 3.1 (Arithmetic Expression Space). *Fix constants $\mu, \lambda \in \mathbb{R}$. An Arithmetic Expression Space (AES) is a pair (M, a) such that:*

- M is a two-dimensional Riemannian manifold with metric g ;
- $a : M \rightarrow \mathbb{R}$ is a smooth scalar field;
- a satisfies the arithmetic flow law, equivalently the eikonal relation

$$\|\nabla a\|_g = \sqrt{\mu^2 + \lambda^2 a^2}. \quad (7)$$

The manifold M provides the geometric stage, the scalar field a records arithmetic assignment, and the parameters μ, λ encode the two elementary propagation intensities. In the next subsection we record the equivalence between the eikonal form and the directional flow description of Section 2.3.

3.2 Compatibility and the eikonal form

The differential law in Section 2.3 is directional: it describes the evolution of a along a curve whose tangent is resolved into additive and multiplicative directions. The eikonal form (7) expresses the same content intrinsically.

Let (M, g) be a Riemannian surface and $a \in C^\infty(M)$. At any point where $\nabla a \neq 0$, the function a determines two distinguished directions: the gradient direction (maximal increase) and the contour direction (level set direction). Along a unit-speed curve $\gamma(s)$ with unit tangent T , one has

$$\frac{d}{ds} a(\gamma(s)) = \langle \nabla a, T \rangle_g.$$

Writing ϕ for the angle between T and the unit gradient direction, this becomes

$$\frac{d}{ds} a(\gamma(s)) = \|\nabla a\|_g \cos \phi.$$

If a satisfies (7), then along such a curve

$$\frac{d}{ds} a(\gamma(s)) = \sqrt{\mu^2 + \lambda^2 a^2} \cos \phi.$$

This reproduces the contour–gradient description and shows that the flow depends only on the metric norm of the gradient and the current assignment value.

The eikonal form also suggests a rectified variable in which propagation becomes linear.

Definition 3.2. Assume $\mu \neq 0$ and define the rectified assignment

$$f := \operatorname{arcsinh}\left(\frac{\lambda a}{\mu}\right).$$

Lemma 3.1. If a satisfies (7) and $\mu \neq 0$, then f satisfies $\|\nabla f\|_g = \lambda$ wherever $\nabla a \neq 0$.

Proof. By the chain rule,

$$\nabla f = \frac{\lambda}{\sqrt{\mu^2 + \lambda^2 a^2}} \nabla a.$$

Taking norms and using (7) gives $\|\nabla f\|_g = \lambda$. □

In particular, when propagation starts from a zero contour and follows a gradient line parameterized by arclength s , one obtains the explicit hyperbolic-sine growth law

$$a(s) = \frac{\mu}{\lambda} \sinh(\lambda s). \quad (8)$$

This rectification will match the geometry of the fundamental model $\mathfrak{E}_0$.

3.3 Model spaces: $\mathfrak{E}_0$ and $\mathfrak{E}_1$

The zeroth kind arithmetic expression space $\mathfrak{E}_0$

Let

$$\mathcal{H} = \{(x, y) \in \mathbb{R}^2 \mid y > 0\}$$

be the upper half-plane. For fixed parameters $\mu > 0$ and $\lambda > 0$, define the metric

$$g_{\mu, \lambda} = \frac{1}{y^2} \left(\frac{dx^2}{\mu^2} + \frac{dy^2}{\lambda^2} \right) \quad (9)$$

and the assignment field

$$a(x, y) = -\frac{x}{y}. \quad (10)$$

Theorem 3.1. *The triple $(\mathcal{H}, g_{\mu, \lambda}, a)$ is an Arithmetic Expression Space in the sense of Definition 3.1. We call it the zeroth kind arithmetic expression space and denote it by $\mathfrak{E}_0 = \mathfrak{E}_0(\mu, \lambda)$.*

Proof. We compute

$$a_x = -\frac{1}{y}, \quad a_y = \frac{x}{y^2} = -\frac{a}{y}.$$

The inverse metric to (9) is

$$g_{\mu, \lambda}^{-1} = \mu^2 y^2 \partial_x^2 + \lambda^2 y^2 \partial_y^2.$$

Therefore

$$\|\nabla a\|_{g_{\mu, \lambda}}^2 = \mu^2 y^2 a_x^2 + \lambda^2 y^2 a_y^2 = \mu^2 + \lambda^2 a^2,$$

which is equivalent to (7). $\square$

The rectified variable $f = \operatorname{arcsinh}(\lambda a / \mu)$ has $\|\nabla f\| = \lambda$ and therefore behaves like a signed distance coordinate measured from the zero locus $a = 0$. In particular, the hyperbolic-sine propagation law (8) becomes visible in $\mathfrak{E}_0$ as growth along gradient geodesics.

The first kind arithmetic expression space $\mathfrak{E}_1$

The zeroth kind space is smooth and its zero locus is an entire geodesic. A singular counterpart collapses the zero locus to an isolated point and thereby introduces the first genuinely nontrivial loops.

Let

$$\mathbb{D} = \{w \in \mathbb{C} \mid |w| < 1\}$$

be the Poincaré disc with the hyperbolic metric of curvature $-\lambda^2$,

$$g_1 = \frac{4}{\lambda^2} \frac{dr^2 + r^2 d\vartheta^2}{(1 - r^2)^2}, \quad (11)$$

where (r, ϑ) are Euclidean polar coordinates. Declare the center $r = 0$ to be the unique zero of the assignment and treat it as a singular point. If s denotes hyperbolic distance from the center, then $a(s) = (\mu/\lambda) \sinh(\lambda s)$ suggests the explicit radial formula

$$a(r) = \frac{\mu}{\lambda} \frac{2r}{1 - r^2}. \quad (12)$$

Away from $r = 0$, this defines an AES on the punctured disc $\mathbb{D} \setminus \{0\}$.

Lemma 3.2. *On $\mathbb{D} \setminus \{0\}$, the pair $(\mathbb{D} \setminus \{0\}, a)$ with metric (11) satisfies the AES eikonal relation (7).*

Proof. Since a is radial, the verification reduces to a one-variable computation in the metric (11); see Appendix A.1. $\square$

Its nontrivial fundamental group makes $\mathfrak{E}_1$ the first natural testing ground for the loop and singularity questions deferred to later work.

3.4 Existence in model settings and remarks on uniqueness

The explicit spaces $\mathfrak{E}_0$ and $\mathfrak{E}_1$ show that the definition is nonempty and that concrete models exist for each choice of parameters $\mu, \lambda > 0$. Beyond these models, the general existence and classification problem is open: one may ask for which Riemannian surfaces (M, g) there exists a nontrivial assignment a solving (7), and to what extent solutions are unique up to geometric equivalence.

The eikonal form also shows that existence is flexible if one allows the metric to vary.

Lemma 3.3 (Local metric rectification). *Let S be a smooth surface with a Riemannian metric g_0 , and let $a \in C^\infty(S)$. If $p \in S$ satisfies $\nabla_{g_0} a(p) \neq 0$, then there exists a neighborhood $U \ni p$ and a conformally rescaled metric g on U such that a satisfies (7) on U .*

Proof. On a surface, locally one may write g_0 in isothermal form $g_0 = e^{2\rho}(du^2 + dv^2)$. Introduce a positive conformal factor α and set $g = \alpha^2 g_0$. Under this rescaling, $\|\nabla_g a\| = \alpha^{-1} \|\nabla_{g_0} a\|$. Choosing

$$\alpha = \frac{\|\nabla_{g_0} a\|}{\sqrt{\mu^2 + \lambda^2 a^2}}$$

forces (7) on U . □

The lemma is intentionally local. Extending such constructions globally requires compatibility across charts and control near critical points and singular loci. These global classification and uniqueness questions will be addressed in later work.

4 The Accumulative Commutative Space and arithmetic torsion

4.1 Arithmetic torsion as commutator defect

Arithmetic torsion is the first invariant measuring that two evaluation histories with the same total additive and multiplicative charges need not agree as arrows in the path groupoid of Section 2.1. In this paper, it is defined as a commutator defect rather than as torsion of an affine connection.

Let

$$\gamma = (g_1, \dots, g_n)$$

be a finite arithmetic path, applied in that order, where each step g_j is either an additive update $g_j = \oplus_{\mu_j} : x \mapsto x + \mu_j$ or a multiplicative update $g_j = \otimes_{\lambda_j} : x \mapsto e^{\lambda_j} x$. The evaluation of γ at an initial value x is

$$\nu_x(\gamma) := g_n \circ \dots \circ g_1(x).$$

We write $\bar{\gamma} = (g_n, \dots, g_1)$ for the reversed history.

Definition 4.1 (Global arithmetic torsion). *The global arithmetic torsion of γ is*

$$\tau_{\text{alg}}(\gamma) := \nu_x(\gamma) - \nu_x(\bar{\gamma}). \quad (13)$$

At first sight (13) depends on the initial value x . The point of the ACS is that this dependence cancels: torsion depends only on the history, not on the basepoint.

4.2 The Accumulative Commutative Space (ACS)

The accumulative commutative space is the plane of total charges. It forgets the current assignment value and remembers only the accumulated additive and multiplicative charges along a history.

Definition 4.2 (ACS path). *The ACS is the plane with coordinates (A, M) , where A records accumulated additive charge and M records accumulated logarithmic multiplicative charge. For each step g_j of γ define*

$$(\Delta A_j, \Delta M_j) = \begin{cases} (\mu_j, 0), & g_j = \oplus_{\mu_j}, \\ (0, \lambda_j), & g_j = \otimes_{\lambda_j}. \end{cases}$$

Starting from $(A_0, M_0) = (0, 0)$, set

$$(A_k, M_k) := \sum_{j=1}^k (\Delta A_j, \Delta M_j), \quad 1 \leq k \leq n.$$

The broken line joining $(A_0, M_0), (A_1, M_1), \dots, (A_n, M_n)$ is the ACS path of γ and is denoted by C_γ .

The endpoint (A_n, M_n) depends only on the total charges and not on their order. Hence C_γ and $C_{\bar{\gamma}}$ always have the same initial and terminal points. When all charges are nonnegative, these two monotone paths bound a canonical region between them. In general we use an oriented filling.

Definition 4.3 (ACS filling). *An ACS filling for γ is an oriented piecewise smooth 2-chain Σ_γ in the (A, M) -plane such that*

$$\partial \Sigma_\gamma = C_{\bar{\gamma}} - C_\gamma \quad (14)$$

as oriented 1-chains.

All integrals in this section are understood in the standard piecewise smooth sense. Changing the parameterization of the edges of C_γ does not change the line integrals, and changing the parameterization of Σ_γ does not change the area integrals.

4.3 Weighted area and boundary formulations

The ACS does not evaluate an expression by itself; it records exactly the commutative data needed to compare two different histories of the same total charge. The basic observation is that evaluation is a weighted boundary integral on the reversed ACS path.

Proposition 4.1 (Evaluation as a weighted edge integral). *For every arithmetic path γ and every initial value x ,*

$$\nu_x(\gamma) = e^{M_\gamma} x + \int_{C_{\bar{\gamma}}} e^M dA, \quad (15)$$

where M_γ is the total multiplicative charge of γ . Consequently, $\tau_{\text{alg}}(\gamma)$ is independent of x .

Proof. Let $I_A \subseteq \{1, \dots, n\}$ be the set of additive indices, and for each $j \in I_A$ define the *future multiplicative charge*

$$M_j^+ := \sum_{\substack{k > j \\ g_k = \otimes \lambda_k}} \lambda_k.$$

When γ is evaluated, the additive contribution μ_j is subsequently scaled by every later multiplicative step, hence by the factor $e^{M_j^+}$. Therefore

$$\nu_x(\gamma) = e^{M_\gamma} x + \sum_{j \in I_A} \mu_j e^{M_j^+}.$$

Now traverse the reversed history $\bar{\gamma}$. The edge corresponding to the original step $g_j = \oplus_{\mu_j}$ is traversed exactly when the current ACS height is M_j^+ , because all multiplicative steps that were later in γ have become earlier in $\bar{\gamma}$. Hence

$$\int_{C_{\bar{\gamma}}} e^M dA = \sum_{j \in I_A} \mu_j e^{M_j^+},$$

with the sign of μ_j incorporated by the orientation of the edge integral. This gives (15).

Applying the same argument to $\bar{\gamma}$ yields

$$\nu_x(\bar{\gamma}) = e^{M_\gamma} x + \int_{C_\gamma} e^M dA,$$

since $M_{\bar{\gamma}} = M_\gamma$. Subtracting cancels the x -term, so $\tau_{\text{alg}}(\gamma)$ is independent of x . $\square$

Two features are worth recording. First, the kernel e^M is forced: every additive step feels all multiplicative steps that come after it. Second, the boundary expression makes torsion a purely commutative quantity on the ACS.

Define the canonical 1-form on the ACS by

$$\eta := e^M dA. \quad (16)$$

Then Proposition 4.1 implies

$$\tau_{\text{alg}}(\gamma) = \int_{C_{\bar{\gamma}}} \eta - \int_{C_\gamma} \eta = \oint_{\partial \Sigma_\gamma} \eta, \quad (17)$$

for any ACS filling Σ_γ .

4.4 The ACS triple identity

Theorem 4.1 (ACS triple identity). *Let $\eta = e^M dA$ be the canonical 1-form on the ACS. For every arithmetic path γ and every ACS filling Σ_γ ,*

$$\tau_{\text{alg}}(\gamma) = \oint_{\partial \Sigma_\gamma} \eta = \iint_{\Sigma_\gamma} d\eta = \iint_{\Sigma_\gamma} e^M dM \wedge dA. \quad (18)$$

In particular, when all charges are nonnegative and Σ_γ is the canonical region between the two monotone ACS paths, $\tau_{\text{alg}}(\gamma)$ is the positive e^M -weighted area between C_γ and $C_{\bar{\gamma}}$.

Proof. Equation (17) gives the boundary integral expression. Since $d\eta = e^M dM \wedge dA$, Stokes' theorem on the plane yields

$$\oint_{\partial \Sigma_\gamma} \eta = \iint_{\Sigma_\gamma} d\eta = \iint_{\Sigma_\gamma} e^M dM \wedge dA,$$

proving (18). $\square$

The right-hand side depends only on the oriented 2-chain Σ_γ with boundary (14); in particular it is invariant under reparameterizations of the boundary paths and of the filling.

From an algebraic viewpoint, the ACS is the commutative shadow of the noncommutative path calculus: the charge map $\gamma \mapsto (A_n, M_n)$ is an abelianization of histories, while the torsion is the defect detected by comparing a word to its reverse.

5 Contact structure and horizontal differential calculus

5.1 The arithmetic expression contact form

The ACS of Section 4.2 records additive and multiplicative charges but forgets the current assignment value. To localize torsion while retaining the assignment, we pass to a three-dimensional space with coordinates (u, v, a) , where u and v play the local additive and multiplicative roles and a records the current assignment.

We work on the manifold

$$\mathcal{C} := \mathbb{R}_{(u,v,a)}^3.$$

Define the 1-forms

$$\omega := \mu du + \lambda a dv, \quad \alpha := da - \omega. \quad (19)$$

For a tangent vector $X = x_u \partial_u + x_v \partial_v + x_a \partial_a$ one computes

$$\alpha(X) = x_a - \mu x_u - \lambda a x_v.$$

Thus $\alpha(X) = 0$ if and only if $x_a = \mu x_u + \lambda a x_v$, which is exactly the infinitesimal propagation law from Proposition 2.1 in Pfaff form.

Proposition 5.1. *If $\mu\lambda \neq 0$, then α is a contact form on $\mathcal{C}$.*

Proof. Since $d\omega = \lambda da \wedge dv$, we have $d\alpha = -\lambda da \wedge dv$ and therefore

$$\alpha \wedge d\alpha = (da - \mu du - \lambda a dv) \wedge (-\lambda da \wedge dv) = \mu\lambda du \wedge da \wedge dv,$$

which is nonzero everywhere when $\mu\lambda \neq 0$. $\square$

The arithmetic form (19) is a coordinatization of the standard local contact model. After rescaling $\tilde{u} = \mu u$ and $\tilde{v} = \lambda v$ and setting $x = \tilde{v}$, $y = a$, $z = a - \tilde{u}$, one obtains $\alpha = dz - y dx$ [1, 2]. The key point is not the existence of a contact structure, but the distinguished horizontal directions singled out by addition and multiplication.

5.2 Horizontal distribution and local non-integrability

The *horizontal distribution* is the rank-2 distribution

$$\mathcal{H} := \ker \alpha \subset T\mathcal{C}.$$

The two basic horizontal lifts of the coordinate fields on the (u, v) -plane are

$$D_u := \partial_u + \mu \partial_a, \quad D_v := \partial_v + \lambda a \partial_a. \quad (20)$$

They satisfy $\alpha(D_u) = \alpha(D_v) = 0$, hence $\mathcal{H} = \text{span}\{D_u, D_v\}$. A horizontal curve $\gamma(s)$ is characterized by $\dot{\gamma}(s) \in \mathcal{H}$, equivalently by the Pfaff constraint $\alpha(\dot{\gamma}) = 0$.

Proposition 5.2. *The horizontal distribution is not integrable. More precisely,*

$$[D_u, D_v] = \mu\lambda \partial_a. \quad (21)$$

Proof. Compute directly:

$$[D_u, D_v] = [\partial_u + \mu \partial_a, \partial_v + \lambda a \partial_a] = \mu\lambda \partial_a. \quad \square$$

Equation (21) is the local differential shadow of arithmetic torsion: an infinitesimal u -move followed by an infinitesimal v -move differs from the reversed order by a purely vertical drift. At finite scale this drift carries the same exponential weighting seen globally in the ACS.

A finite commutator square

Let Φ_u^h and Φ_v^k denote the time- h and time- k flows of D_u and D_v . They are

$$\Phi_u^h(u, v, a) = (u + h, v, a + \mu h), \quad \Phi_v^k(u, v, a) = (u, v + k, e^{\lambda k} a).$$

Comparing the two two-step histories with the same base endpoint,

$$p_{uv} := \Phi_v^k \circ \Phi_u^h(u, v, a), \quad p_{vu} := \Phi_u^h \circ \Phi_v^k(u, v, a),$$

one finds

$$a(p_{uv}) - a(p_{vu}) = \mu h(e^{\lambda k} - 1), \quad (22)$$

whose first-order part is $\mu\lambda h k$.

5.3 The horizontal differential δ

The horizontal distribution comes with a natural differential operator obtained by projecting the de Rham differential to the horizontal directions.

Definition 5.1. For a smooth scalar field $F(u, v, a)$ on $\mathcal{C}$, its horizontal differential is

$$\delta F := (D_u F) du + (D_v F) dv. \quad (23)$$

Lemma 5.1. For any smooth F ,

$$\delta F = dF - (\partial_a F)\alpha. \quad (24)$$

In particular, $\delta a = \omega$ and $\delta u = du$, $\delta v = dv$.

Proof. Expand $dF = (\partial_u F) du + (\partial_v F) dv + (\partial_a F) da$ and use $da = \alpha + \omega$ with (19). Then

$$dF - (\partial_a F)\alpha = (\partial_u F + \mu \partial_a F) du + (\partial_v F + \lambda a \partial_a F) dv = (D_u F) du + (D_v F) dv.$$

The special cases follow by taking $F = a, u, v$. □

The operator δ is not nilpotent: its failure to square to zero is precisely the commutator curvature of the horizontal distribution.

Proposition 5.3. For any smooth F ,

$$\delta^2 F = \mu \lambda (\partial_a F) du \wedge dv. \quad (25)$$

In particular, $\delta^2 a = \mu \lambda du \wedge dv$.

Proof. By definition, $\delta^2 F = ([D_u, D_v]F) du \wedge dv$. Using (21) gives (25). □

This local 2-form density $\mu \lambda du \wedge dv$ will later appear as the curvature term in the horizontal holomorphic and twisted harmonic operators.

5.4 Degenerate limits and the singular model

The contact condition in Proposition 5.1 fails when one of the propagation intensities vanishes. This is consistent with arithmetic intuition: if one generator is missing, the commutator defect disappears.

The additive-only limit $\lambda = 0$

If $\lambda = 0$, then $\alpha = da - \mu du$ and $d\alpha = 0$, so $\alpha \wedge d\alpha = 0$. The horizontal distribution is spanned by ∂_v and $\partial_u + \mu \partial_a$ and is integrable. In this limit, $D_v = \partial_v$ and $[D_u, D_v] = 0$, hence $\delta^2 = 0$.

The multiplicative-only limit $\mu = 0$

If $\mu = 0$, then $\alpha = da - \lambda a dv$ and again $\alpha \wedge d\alpha = 0$. The horizontal distribution is spanned by ∂_u and $\partial_v + \lambda a \partial_a$ and is integrable. Again $[D_u, D_v] = 0$ and $\delta^2 = 0$.

These degenerate limits are precontact/foliated structures rather than contact structures. They reflect the purely additive or purely multiplicative regimes where torsion vanishes. This degeneration should not be confused with the singular AES model $\mathfrak{E}_1$ of Section 3.3, where $\mu \lambda \neq 0$ but the assignment has a geometric singularity at its zero locus.

6 Horizontal holomorphic fields and twisted harmonicity

6.1 Horizontal holomorphic fields

Section 5 canonically associates to arithmetic propagation a contact manifold $\mathcal{C} = \mathbb{R}_{(u,v,a)}^3$ with horizontal distribution $\mathcal{H} = \ker \alpha$ spanned by the two distinguished horizontal fields D_u and D_v . In this section we extract from $(\mathcal{H}, D_u, D_v)$ a first-order analytic operator package. The emphasis is on what is forced by the horizontal geometry, not on importing the full classical function theory of complex analysis.

We work with complex-valued fields $F : \mathcal{C} \rightarrow \mathbb{C}$ written as $F = f + ig$ with f, g real-valued.

Definition 6.1 (Horizontal holomorphic fields). *A complex-valued field $F = f + ig$ on $\mathcal{C}$ is horizontally holomorphic if it satisfies the arithmetic Cauchy–Riemann system*

$$D_u f = D_v g, \quad D_v f = -D_u g. \quad (26)$$

This is the direct horizontal analogue of the classical Cauchy–Riemann equations: it equates the horizontal derivatives of f and g along the distinguished arithmetic directions.

Lemma 6.1 (Horizontal conformality). *If $F = f + ig$ is horizontally holomorphic, then the two horizontal derivative vectors $(D_u f, D_u g)$ and $(D_v f, D_v g)$ are orthogonal and have the same length. Moreover,*

$$J_H(F) := D_u f D_v g - D_v f D_u g = (D_u f)^2 + (D_v f)^2 = (D_u g)^2 + (D_v g)^2, \quad (27)$$

and

$$\delta f \wedge \delta g = J_H(F) du \wedge dv,$$

where δ is the horizontal differential of Section 5.3.

Proof. The orthogonality and equal-length statements follow immediately from (26). For the Jacobian identity, substitute (26) into the definition of $J_H(F)$ to obtain $J_H(F) = (D_u f)^2 + (D_v f)^2$, and similarly $J_H(F) = (D_u g)^2 + (D_v g)^2$. The wedge identity is a direct expansion:

$$\delta f \wedge \delta g = (D_u f du + D_v f dv) \wedge (D_u g du + D_v g dv) = (D_u f D_v g - D_v f D_u g) du \wedge dv.$$

□

As a basic example, the field $F(u, v, a) = u + iv$ is horizontally holomorphic since $D_u u = 1$, $D_v v = 1$, and the mixed derivatives vanish.

6.2 Arithmetic Cauchy–Riemann operator package

Define the first-order operators on complex-valued fields by

$$\bar{\partial}_{\text{AEG}} := \frac{1}{2}(D_u + iD_v), \quad \partial_{\text{AEG}} := \frac{1}{2}(D_u - iD_v). \quad (28)$$

Then Definition 6.1 is equivalently expressed as

$$\bar{\partial}_{\text{AEG}} F = 0. \quad (29)$$

The horizontal calculus provides a convenient differential-form presentation of (28). Let

$$\delta F = (D_u F) du + (D_v F) dv.$$

Writing $\zeta := du + i dv$ and $\bar{\zeta} := du - i dv$, one has the decomposition

$$\delta F = (\partial_{\text{AEG}} F) \zeta + (\bar{\partial}_{\text{AEG}} F) \bar{\zeta}. \quad (30)$$

In particular, F is horizontally holomorphic if and only if δF is of type $(1, 0)$ with respect to the horizontal splitting determined by D_u and D_v .

6.3 Twisted harmonicity

The operators (28) admit a second-order factorization whose defect is exactly the commutator curvature of the horizontal distribution.

Definition 6.2 (Horizontal Laplacian). *Define the horizontal Laplacian on scalar fields by*

$$\Delta_H := D_u^2 + D_v^2. \quad (31)$$

Proposition 6.1 (Factorization). *On complex-valued fields F ,*

$$4 \partial_{\text{AEG}} \bar{\partial}_{\text{AEG}} F = (\Delta_H + i\mu\lambda \partial_a) F, \quad 4 \bar{\partial}_{\text{AEG}} \partial_{\text{AEG}} F = (\Delta_H - i\mu\lambda \partial_a) F. \quad (32)$$

Proof. Expand

$$4 \partial_{\text{AEG}} \bar{\partial}_{\text{AEG}} F = (D_u - iD_v)(D_u + iD_v)F = (D_u^2 + D_v^2)F + i(D_u D_v - D_v D_u)F.$$

Using the bracket identity $[D_u, D_v] = \mu\lambda \partial_a$ from Proposition 5.2 gives the first formula. The second is analogous. $\square$

Theorem 6.1 (Twisted harmonicity). *If F is horizontally holomorphic, then it satisfies the twisted second-order equation*

$$(\Delta_H + i\mu\lambda \partial_a) F = 0. \quad (33)$$

Equivalently, writing $F = f + ig$,

$$\Delta_H f = \mu\lambda \partial_a g, \quad \Delta_H g = -\mu\lambda \partial_a f. \quad (34)$$

Proof. If $\bar{\partial}_{\text{AEG}} F = 0$, then applying ∂_{AEG} and using Proposition 6.1 yields (33). The component form follows by taking real and imaginary parts. $\square$

Equation (33) exhibits the precise role of local torsion: the twisting term is the constant commutator density $\mu\lambda$ which already appeared as $\delta^2 a$ in Proposition 5.3.

6.4 Scope and limitations

The present section introduces a first-order operator package determined by the horizontal distribution and records its basic second-order consequence. We do not develop a global function theory parallel to classical complex analysis. In particular, we do not address boundary integral kernels, global boundary value problems, analytic continuation, or a full functional-analytic theory of solution spaces. These belong to later work building on the operators defined here.

7 Invariance and mathematical positioning

7.1 Normalization and parameter rescaling

The constructions introduced so far carry two numerical parameters μ and λ recording the additive and multiplicative propagation intensities. This section records the basic normalizations and invariances showing which combinations are geometric and which are choices of units.

Natural units

On the contact manifold $\mathcal{C} = \mathbb{R}_{(u,v,a)}^3$ we introduce the dimensionless variables

$$\tilde{u} := \mu u, \quad \tilde{v} := \lambda v. \quad (35)$$

In these variables the contact form (19) becomes

$$\alpha = da - d\tilde{u} - a d\tilde{v}, \quad (36)$$

and the horizontal frame (20) becomes

$$D_u = \mu(\partial_{\tilde{u}} + \partial_a), \quad D_v = \lambda(\partial_{\tilde{v}} + a\partial_a). \quad (37)$$

Thus the μ - and λ -dependence is a pure scaling of the preferred horizontal directions.

The fundamental curvature density is likewise unit-free:

$$\mu\lambda du \wedge dv = d\tilde{u} \wedge d\tilde{v}. \quad (38)$$

This is the geometric meaning of the constant $\mu\lambda$ appearing in $[D_u, D_v] = \mu\lambda\partial_a$ and in the twisted harmonicity term $(\Delta_H + i\mu\lambda\partial_a)$.

Change of logarithmic base

The multiplicative coordinate v is a logarithmic parameter. Choosing a different logarithmic base is equivalent to rescaling v and λ while keeping the product $M = \lambda v$ fixed. Concretely, if multiplicative propagation is written as

$$x \mapsto b^{\lambda v} x,$$

then $\lambda = \lambda_b \log b$ in the conventions used here. Equivalently, one may replace (λ, v) by $(c\lambda, v/c)$ for any $c > 0$ without changing the multiplicative factors $e^{\lambda v}$ that appear in the generator calculus and in the ACS weight e^M .

7.2 Coordinate covariance of the flow and contact form

The flow law admits equivalent coordinate-free formulations, and the local analytic operators are covariant under the normalizations above.

Pfaff and horizontal formulations

On $\mathcal{C}$ the Pfaff equation $\alpha = 0$ is equivalent to the propagation law: a curve $\gamma(s)$ is horizontal if and only if its tangent satisfies $\alpha(\dot{\gamma}) = 0$, i.e. $da = \mu du + \lambda a dv$ along γ . This is the local, coordinate-free content of Proposition 2.1.

The horizontal differential δ is similarly intrinsic: for a scalar field F ,

$$\delta F = dF - (\partial_a F)\alpha,$$

so $\delta F(X) = dF(X)$ for every horizontal vector $X \in \ker \alpha$. In particular, the definitions of $\bar{\partial}_{\text{AEG}}$ and ∂_{AEG} depend only on the distinguished ordered frame (D_u, D_v) of the horizontal distribution.

Covariance under unit rescaling

Under the natural-unit change (35), the operators scale as

$$\bar{\partial}_{\text{AEG}} = \frac{1}{2}(D_u + iD_v) = \frac{\mu}{2}(\partial_{\tilde{u}} + \partial_a) + \frac{i\lambda}{2}(\partial_{\tilde{v}} + a\partial_a),$$

and similarly for ∂_{AEG} . The factorization identity

$$4\partial_{\text{AEG}}\bar{\partial}_{\text{AEG}} = \Delta_H + i\mu\lambda\partial_a$$

is therefore invariant in form: the only scalar datum it involves is the curvature density $\mu\lambda$, which becomes the unit-free density (38) in natural units.

7.3 Invariance properties of ACS quantities

The ACS package is a global commutative shadow of the local contact geometry. Its basic objects depend only on charges and not on parameterizations.

Reparametrization invariance

The ACS path C_γ is a piecewise linear 1-chain in the (A, M) -plane. Any reparametrization of its edges preserves the line integrals of the 1-form $\eta = e^M dA$. Likewise, an ACS filling Σ_γ is an oriented 2-chain with boundary $C_{\bar{\gamma}} - C_\gamma$, and any reparametrization preserves the area integral of $d\eta = e^M dM \wedge dA$. This is the analytic content behind the statement that torsion depends only on the history, not on how one traverses its ACS shadow.

Normalization of the weight

The exponential weight e^M is forced by propagation: each additive increment is scaled by every later multiplicative increment. Writing the multiplicative charge coordinate as

$$M = \sum \lambda_j$$

means we have chosen the natural exponential coordinate on the multiplicative group. A change of logarithmic base is absorbed by rescaling the pair (λ, v) while keeping $M = \lambda v$ invariant, and therefore does not change the weight in the triple identity.

7.4 Positioning relative to contact/CR/sub-Riemannian geometry and rewriting

The foundational structures of AEG sit naturally among several established geometric languages. The point is not to rename the theory into an existing one, but to clarify what is standard and what is arithmetic-specific.

Contact viewpoint and jet-space form

The Pfaff form $\alpha = da - \mu du - \lambda a dv$ is of the standard contact type $dz - p dx - q dy$. The Darboux reduction $\alpha = dz - y dx$ shows that, locally, the contact structure is not new; what is new is the arithmetic meaning of the preferred horizontal frame (D_u, D_v) singled out by addition and multiplication.

Horizontal/CR viewpoint

The operators $\bar{\partial}_{\text{AEG}}$ and ∂_{AEG} are CR-type first-order operators built from an oriented horizontal frame. The associated second-order operator $\Delta_H = D_u^2 + D_v^2$ is a horizontal Laplacian, and the twisting term $\pm i\mu\lambda \partial_a$ is the explicit curvature correction forced by non-integrability.

Rewriting and groupoids

Section 2.1 packages evaluation histories as arrows in a path groupoid. The ACS is an abelianization of this noncommutative calculus, keeping only total charges. The torsion and its boundary/area formulas quantify precisely what is lost in passing to this commutative shadow. This is the organizing principle behind the equality ladder: endpoints may coincide while arrows remain distinct, and loops encode relations among histories.

8 Outlook

8.1 Classification problems for AES

The definition of an Arithmetic Expression Space (Definition 3.1) is intentionally minimal. Beyond the explicit constructions $\mathfrak{E}_0$ and $\mathfrak{E}_1$, it is natural to ask how canonical these models are and what the correct notion of equivalence should be.

1. **Existence with fixed background.** For a given Riemannian surface (M, g) and fixed parameters (μ, λ) , under what geometric hypotheses does there exist a nonconstant assignment a satisfying the eikonal law (7)? How does the answer change if one allows prescribed singularities (as in $\mathfrak{E}_1$)?
2. **Uniqueness up to equivalence.** If (M, g, a) and (M', g', a') are AES with the same parameters, what is the correct equivalence relation: isometry, conformal equivalence, or a weaker “unit-change” relation combining coordinate rescalings with parameter rescalings? Under such an equivalence, when are solutions unique?
3. **Normalization targets.** The contact geometry admits natural units (35) and a Darboux reduction. On the AES side, one seeks analogous normalization statements: for example, in what sense can (μ, λ) be normalized, and which quantities (e.g. curvature scale) remain as genuine moduli?
4. **Model rigidity.** To what extent is $\mathfrak{E}_0$ characterized by the existence of an assignment simultaneously satisfying an eikonal law and a Laplace-eigenfunction relation? Is there a classification of AES models on constant-curvature backgrounds?

Even partial answers in restricted model classes would strengthen the foundational layer by clarifying what is canonical and what is a choice.

8.2 Loop and relation theories

This paper has focused on *paths*: evaluation histories as arrows in a path groupoid. The next structural layer should focus on *relations*: closed words and their residues across different levels of forgetting. The basic primitive is not equality alone, but *neutrality*: the sense in which a nontrivial history returns to a null effect.

1. **Four neutralities.** For a loop based at x , one may ask for value neutrality (same endpoint value), operator neutrality (the induced transformation is the identity on a domain), charge neutrality (its ACS endpoint is $(0, 0)$), and torsion neutrality (its ACS weighted area vanishes). These notions are distinct and should be separated by explicit examples.
2. **Equality ladder via groupoids.** A natural organizing principle is to compare histories as arrows (strong equality), then mod out by chosen relations among loops (intermediate equalities), and finally forget arrows and retain only endpoints (weak equality). One goal is to define a minimal set of relations which captures common algebraic identities while preserving torsion as a meaningful obstruction.
3. **Holonomy and torsion detection.** The ACS triple identity already exhibits torsion as an area-type invariant of a commutative shadow. The contact picture exhibits a vertical holonomy for horizontal commutator loops. A central question is to unify these into a loop invariant that detects when a relation is invisible at the level of charges but still visible through torsion or curvature.
4. **Singular backgrounds.** The punctured model $\mathfrak{E}_1$ suggests that singular loci may generate nontrivial monodromy and new neutrality phenomena. Understanding which invariants survive and which are obstructed by singularities is a natural bridge between relation theory and analysis.

8.3 Further analytic directions

Section 6 establishes a first-order operator package forced by the horizontal distribution and records its twisted second-order consequence. A realistic next analytic stage is to develop a small suite of results on the basic backgrounds before attempting a global theory.

1. **Canonical second-order operators.** Besides $\Delta_H = D_u^2 + D_v^2$, one may seek the correct self-adjoint horizontal Laplacian and sub-Laplacian variants compatible with the contact volume form, together with their Green-type identities on model domains.
2. **Explicit solution families.** Construct nontrivial families of horizontally holomorphic fields and twisted harmonic fields on the basic local model $\mathcal{C}$ and on the model AES backgrounds. Identify which families are stable under the unit rescalings of Section 7.1.

3. **Local solvability and normal forms.** Establish a local normal form for the arithmetic Cauchy–Riemann system and identify obstructions coming from the curvature density $\mu\lambda$. Determine when one can solve $\partial_{\text{AEG}} F = G$ locally with prescribed regularity.
4. **Boundary and kernel problems.** Formulate the first boundary value problems in model settings and identify whether there exist natural integral kernels adapted to the horizontal operators. Any such development should keep the arithmetic interpretation of D_u and D_v visible and should avoid becoming a generic theory detached from propagation.

References

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A Supplementary calculations for AES and ACS

A.1 Model calculations for $\mathfrak{E}_0$ and $\mathfrak{E}_1$

Verification for $\mathfrak{E}_1$

We verify Lemma 3.2. Recall the hyperbolic metric of curvature $-\lambda^2$ on the Poincaré disc,

$$g_1 = \frac{4}{\lambda^2} \frac{dr^2 + r^2 d\vartheta^2}{(1 - r^2)^2},$$

and the radial assignment

$$a(r) = \frac{\mu}{\lambda} \frac{2r}{1 - r^2}.$$

Since a is independent of ϑ , we have $da = a_r dr$ with

$$a_r = \frac{2\mu}{\lambda} \frac{1 + r^2}{(1 - r^2)^2}.$$

The inverse metric is

$$g_1^{-1} = \frac{\lambda^2}{4} (1 - r^2)^2 \left(\partial_r^2 + \frac{1}{r^2} \partial_\vartheta^2 \right),$$

so

$$\|\nabla a\|_{g_1}^2 = g_1^{-1}(da, da) = \frac{\lambda^2}{4} (1 - r^2)^2 a_r^2 = \mu^2 \frac{(1 + r^2)^2}{(1 - r^2)^2}.$$

On the other hand,

$$\mu^2 + \lambda^2 a^2 = \mu^2 + \mu^2 \frac{4r^2}{(1 - r^2)^2} = \mu^2 \frac{(1 - r^2)^2 + 4r^2}{(1 - r^2)^2} = \mu^2 \frac{(1 + r^2)^2}{(1 - r^2)^2}.$$

Thus $\|\nabla a\|_{g_1}^2 = \mu^2 + \lambda^2 a^2$, which is equivalent to the eikonal relation (7) on $\mathbb{D} \setminus \{0\}$.

A.2 ACS weight, area form, and boundary conventions

We record the sign and orientation conventions implicit in Section 4.4.

The ACS is the (A, M) -plane equipped with its standard orientation. We write the oriented area form as

$$dM \wedge dA.$$

With this convention, the canonical 1-form

$$\eta = e^M dA$$

satisfies

$$d\eta = e^M dM \wedge dA.$$

If Σ is an oriented piecewise smooth 2-chain in the plane, its boundary $\partial\Sigma$ is oriented by the “outward normal first” convention. Stokes’ theorem then reads

$$\oint_{\partial\Sigma} \eta = \iint_{\Sigma} d\eta.$$

For the ACS triple identity, the required filling condition is $\partial\Sigma_\gamma = C_{\bar{\gamma}} - C_\gamma$ as oriented 1-chains. Any reparameterization of the piecewise linear boundary paths preserves the value of the boundary integral.

B Supplementary calculations for contact, δ , and holomorphicity

B.1 Contact computations and bracket identities

We collect the computations used in Section 5.1–5.2.

Contact condition

With $\omega = \mu du + \lambda a dv$ and $\alpha = da - \omega$, we have

$$d\omega = \lambda da \wedge dv, \quad d\alpha = -\lambda da \wedge dv.$$

Therefore

$$\alpha \wedge d\alpha = (da - \mu du - \lambda a dv) \wedge (-\lambda da \wedge dv) = \mu\lambda du \wedge da \wedge dv.$$

In particular, α is a contact form if and only if $\mu\lambda \neq 0$.

Horizontal frame and bracket

The horizontal lifts $D_u = \partial_u + \mu\partial_a$ and $D_v = \partial_v + \lambda a\partial_a$ satisfy $\alpha(D_u) = \alpha(D_v) = 0$. Their bracket is

$$[D_u, D_v] = [\partial_u + \mu\partial_a, \partial_v + \lambda a\partial_a] = \mu\lambda\partial_a.$$

Darboux coordinates

Rescale $\tilde{u} = \mu u$ and $\tilde{v} = \lambda v$, and set $x = \tilde{v}$, $y = a$, $z = a - \tilde{u}$. Then $dz = da - \mu du$ and $dx = \lambda dv$, so

$$\alpha = da - \mu du - \lambda a dv = dz - y dx,$$

which is the Darboux normal form.

B.2 Details of the δ -calculus

For a smooth scalar field $F(u, v, a)$,

$$\delta F = (D_u F) du + (D_v F) dv.$$

Using $da = \alpha + \omega$ with $\omega = \mu du + \lambda a dv$, we obtain

$$dF - (\partial_a F)\alpha = (\partial_u F + \mu\partial_a F) du + (\partial_v F + \lambda a\partial_a F) dv = (D_u F) du + (D_v F) dv = \delta F.$$

Moreover,

$$\delta^2 F = ([D_u, D_v]F) du \wedge dv = \mu\lambda(\partial_a F) du \wedge dv.$$

B.3 Model holomorphicity computations

We record the algebra underlying Proposition 6.1 and Theorem 6.1.

For complex-valued F , define

$$\bar{\partial}_{\text{AEG}} = \frac{1}{2}(D_u + iD_v), \quad \partial_{\text{AEG}} = \frac{1}{2}(D_u - iD_v).$$

Then

$$4\partial_{\text{AEG}}\bar{\partial}_{\text{AEG}}F = (D_u - iD_v)(D_u + iD_v)F = (D_u^2 + D_v^2)F + i(D_u D_v - D_v D_u)F.$$

Using $[D_u, D_v] = \mu\lambda\partial_a$, we obtain

$$4\partial_{\text{AEG}}\bar{\partial}_{\text{AEG}}F = (\Delta_H + i\mu\lambda\partial_a)F, \quad \Delta_H = D_u^2 + D_v^2.$$

Similarly,

$$4\bar{\partial}_{\text{AEG}}\partial_{\text{AEG}}F = (\Delta_H - i\mu\lambda\partial_a)F.$$

If F is horizontally holomorphic, i.e. $\bar{\partial}_{\text{AEG}}F = 0$, then applying ∂_{AEG} yields

$$(\Delta_H + i\mu\lambda\partial_a)F = 0,$$

which is the twisted harmonicity equation.

C Supplementary examples on equality levels and first neutralities

C.1 Examples separating equality levels

C.2 First neutralities and preliminary loop phenomena